AI Spring 2026
Quiz 2

Total Marks: 100 Total Time: 50 Minutes

Instructions: Use your **4-digit Student ID = 23K - A B C D**.
Use **1 = Yes**, **0 = No**.

Dataset Generation

(a) Compute the following:

- $T_1 = (A + 2B) mod 3$
- $T_2 = (C + D) mod 3$
- $T_3 = (A + C + D) mod 2$
- $T_4 = (B + D) mod 2$

Convert any value of **2 → 1** (i.e., non-zero = 1).

(b) Complete the dataset:

Ex	Outlook	Wind	Humidity	Play
1	Sunny	Weak	High	T_1
2	Sunny	Strong	Normal	T_2
3	Overcast	Weak	High	1
4	Overcast	Strong	Normal	1
5	Rain	Weak	High	T_3
6	Rain	Strong	Normal	T_4
7	Sunny	Weak	Normal	0
8	Rain	Weak	Normal	0

Using the **ID3 decision tree algorithm *Show all calculations***:

1. Compute entropy $H(Play)$
2. Compute information gain for **Outlook, Wind, Humidity**
3. Select the root node
4. Construct the final decision tree (show key steps)
5. Classify using your tree: Outlook = Sunny, Wind = Strong, Humidity = High